MONIL PANCHAL

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Summary

Results-driven AI/ML, Backend, and Data Analytics Engineer with 1–3 years of experience building scalable data-driven applications and intelligent systems. Proficient in Python, FastAPI, Django, SQL, and data visualization, with a strong foundation in machine learning and statistical analysis. Experienced in developing robust backend systems, RESTful APIs, and efficient data pipelines. Skilled in applying AI and data analytics to automate workflows, extract insights, and support data-driven decision-making.

Skills

- **Languages & Querying:** Python, SQL (MySQL, PostgreSQL), JavaScript, TypeScript
- **Backend :** FastAPI, Django, REST APIs, API Integrations
- **Frontend:** React.js, HTML, CSS
- **Data & ML:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, SciPy, NLTK
- **Visualization Tools:** Power BI, Tableau
- **Cloud & Big Data:** AWS (S3, EC2, lambda)
- **Analytical Skills:** A/B Testing, Regression Analysis, Statistical Modelling, ETL, Data Wrangling
- **Artificial Intelligence & NLP:** Natural Language Processing (NLP), Transformers, LLM Integration (OpenAI/Claude APIs), Prompt Engineering, Text Classification

Experience

Data Analyst **Freelancing** *Remote, India* 2025 - Current

- Delivered custom data analytics and reporting solutions for small and mid-sized businesses across industries.
- Collected, cleaned, and transformed raw data from Excel, CSV, and databases to generate actionable business insights.
- Performed data analysis and trend identification to help businesses improve performance and identify growth opportunities.
- Collaborated with stakeholders to understand business requirements and deliver tailored analytics solutions.
- Provided data cleaning, validation, and structuring services, improving reliability of client datasets. Supported clients in ad-hoc analysis and reporting needs, ensuring timely and accurate insights.

Data Support Executive **Advantmed India LLP** *Ahmedabad* 01/2024 - 03/2024

- Identified patients requiring regular check-ups based on data patterns, supporting preventive care initiatives to reduce high-cost claims. Managed healthcare insurance data including claims, policies, and patient records using Excel and SQL
- Wrote SQL queries to extract and transform data from databases for reporting purposes.
- Assisted internal teams by providing structured reports and data insights for better decision-making. Performed data cleaning and validation in Excel to ensure consistency and reliability of insurance data
- Supported cost-optimization efforts by enabling early intervention strategies through data-driven decision-making
- Assigned and optimized visit routes for Nurse Practitioners, ensuring efficient scheduling across multiple locations to reduce travel time and improve productivity.

Python Developer (AI/ML) Intern **Brainer Hub** *Ahmedabad* 07/2024 - 02/2025

- Built and optimized SQL queries to extract, clean, and transform raw data from multiple sources into structured formats for analysis.
- Developed and maintained backend systems using Python, delivering scalable solutions for multiple client projects.
- Created data visualizations and executive-level reports using Matplotlib and Seaborn, communicating insights to non-technical stakeholders.
- Built and deployed RESTful APIs (FastAPI/Flask) to handle data processing, integrations, and business logic.
- Performed bug fixing, performance tuning, and code optimization, improving system reliability and response time.
- Collaborated with frontend and business teams to deliver end-to-end client solutions based on requirements.

Projects

Geely Automotive Pricing Model

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Statsmodels

- Built a multiple linear regression model to predict car prices using real-world automobile dataset, identifying key factors influencing pricing.
- Performed data preprocessing, feature engineering, and encoding of categorical variables to improve model performance.
- Conducted EDA and correlation analysis (heatmaps, statistical summaries) to uncover relationships between vehicle attributes and price. Applied feature selection techniques (RFE, VIF) to remove multicollinearity and optimize model accuracy.
- Provided actionable insights for pricing strategy, helping simulate how design and specifications impact vehicle pricing.

Movie Recommender System

Python, FastAPI, React/Frontend, Pandas, Numpy, Scikit-learn Machine Learning

- Designed and developed a full-stack movie recommendation system with a custom frontend and backend architecture.
- Built RESTful APIs using FastAPI to handle recommendation requests and serve model predictions efficiently.
- Implemented a content-based recommendation engine using cosine similarity on movie features such as genres, keywords, and metadata.
- Developed a dynamic and responsive frontend interface (React/JavaScript) for seamless user interaction and real-time recommendations. Optimized backend performance using precomputed similarity matrices and efficient data structures for faster response time.
- Structured the system following a modular architecture (frontend + backend + ML layer) for scalability and maintainability. Handled data preprocessing, feature engineering, and model building using Pandas, NumPy, and Scikit-learn.

Lead Scoring Case Study

Python, scikit-learn, Logistic Regression, EDA

- Developed a machine learning lead-scoring model for an EdTech firm to predict conversion probability of prospective customers.
- Performed comprehensive EDA on 9,000+ leads, handling missing values, outliers, and feature engineering to improve model accuracy.
- Achieved AUC-ROC of 0.88, enabling the sales team to prioritize high-intent leads and improve conversion rate by 30%.

Education

International Institute of Information Technology Bangalore (IIITB) 2022 - 2023

Post Graduate Diploma in Data Science

- Bangalore/Hybrid | CGPA: 3.57 / 4.0

L.J. Institute of Engineering and Technology (LJIET) 2016 - 2021

Bachelor of Engineering (B.E)

- Ahmedabad, Gujarat | CGPA: 8.03/10

Certifications

Python and SQL Certification 2022

- DataCamp / HackerRank / CodeChef

The Fundamentals of Digital Marketing 2024

- Google

Data Analytics and Visualization Virtual Experience 2023

- Accenture